



用笔记本电脑 运行大语言模型

Miley Fu @mileyfu WasmEdge DevRel, CNCF Ambassador

Michael Yuan @juntao WasmEdge Founder, Author of 5 Books on Software Development

<https://github.com/WasmEdge/WasmEdge>

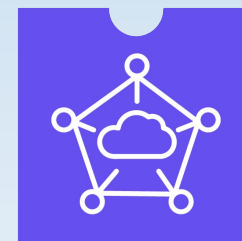




WasmEdge 简介

Wasm用在服务端

About WasmEdge



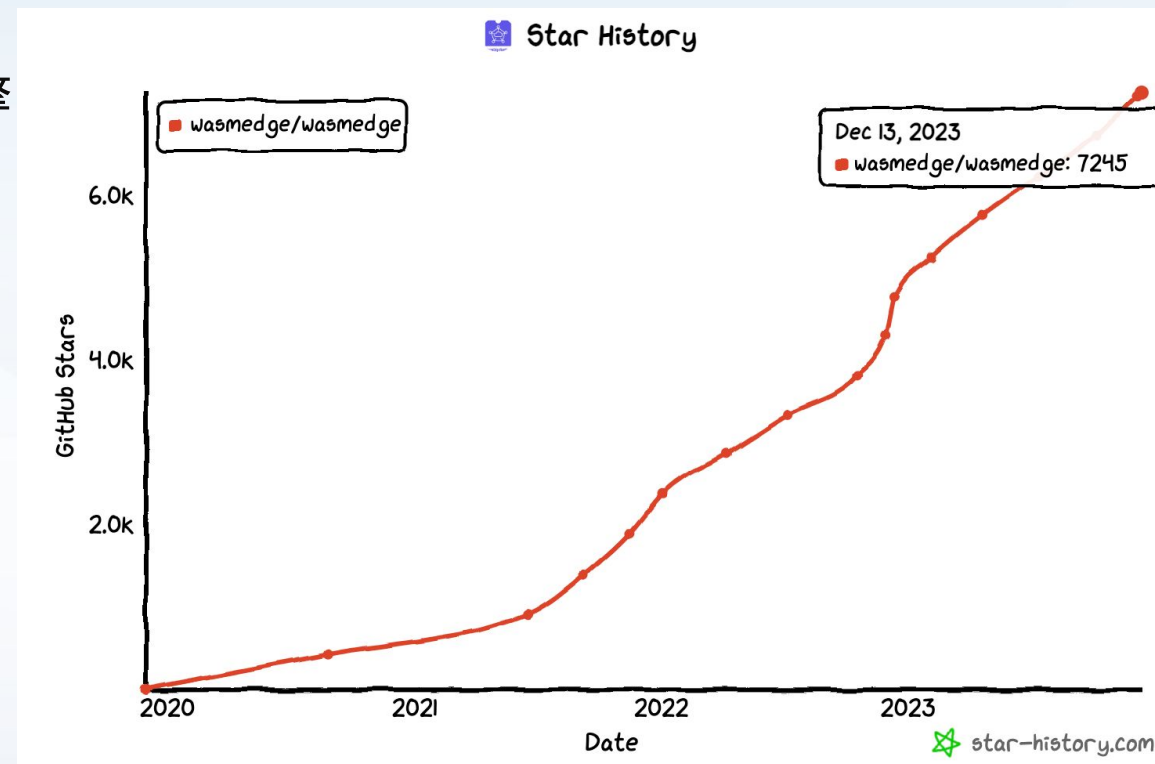
WasmEdgeRuntime

- A WebAssembly (Wasm) runtime.
- Highly efficient, lightweight and safe virtualization solution.
 - Web browsers, IoT 设备, 边缘场景
 - cloud data centers+公链智能合约执行引擎
- 开源社区贡献者活跃: 150+

The No.1 WebAssembly runtime among CNCF projects. Co-launched Wasm landscape



CLOUD NATIVE SANDBOX





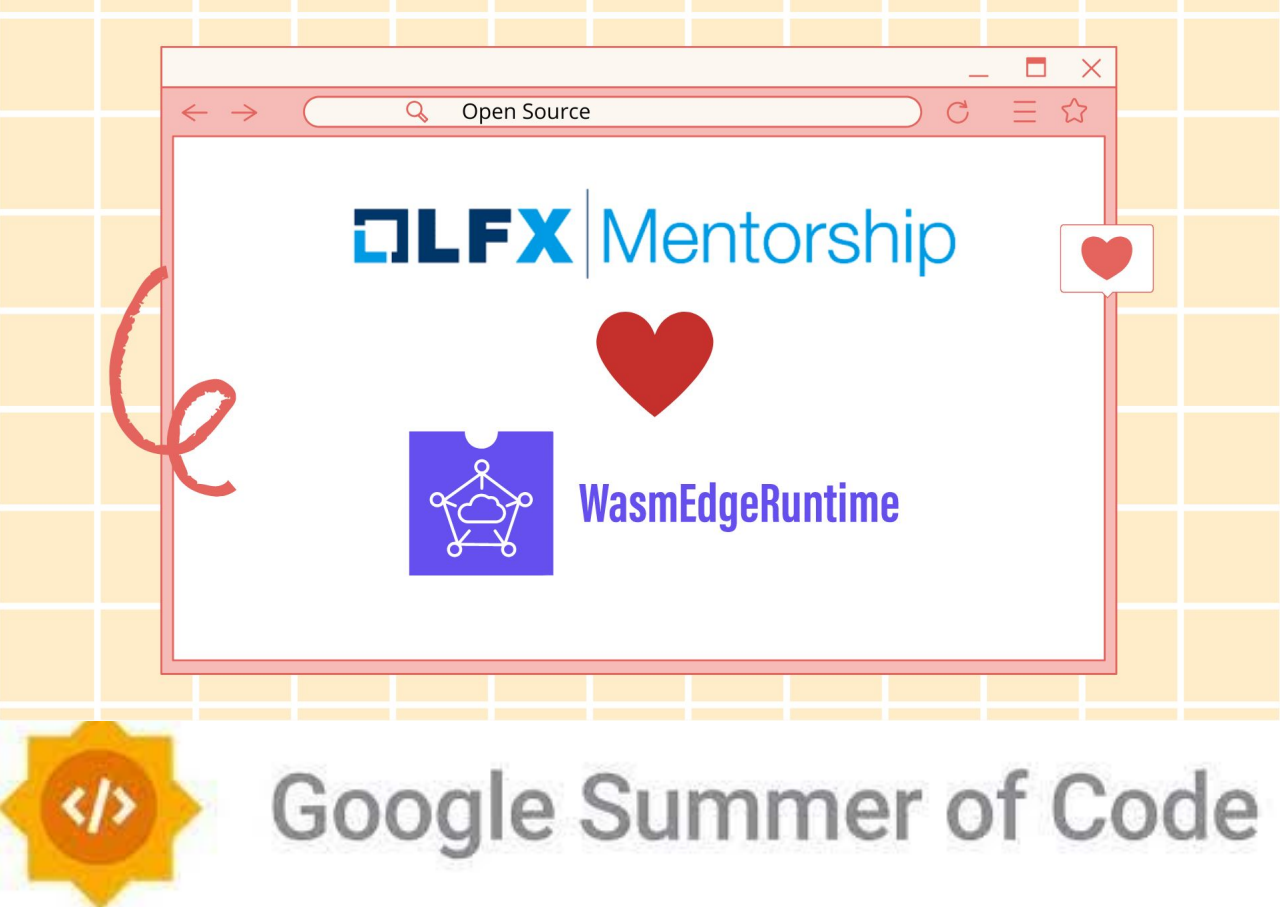
开源之夏2023明天开启报名！欢迎报名 WasmEdge 社区项目

 **WasmEdge**
CNCF 沙箱项目，为边缘优化的Wasm虚拟机

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一年一度的开源之夏又开始啦！明天（4月29日）起，在校学生可以登录开源之夏



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#3063 opened 2 weeks ago by XinyuShe			
bug: fd_pwrite after setting fdflags::append writes from the wrong offset	bug c-WASI good first issue		
platform-linux			
#3062 opened 2 weeks ago by yagehu			
Print hard link file number error.	bug c-WASI good first issue		platform-windows
#3054 opened 3 weeks ago by Userzxcvbnm			
Fail to set file time ?	bug c-WASI good first issue		platform-windows
#3052 opened 3 weeks ago by Userzxcvbnm			
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Edge/Bare metal



AI/Machine Learning



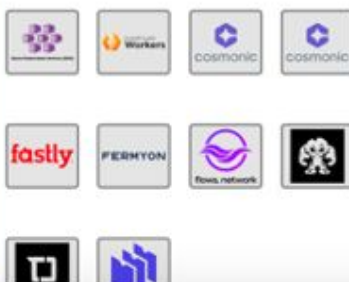
Embedded Functions



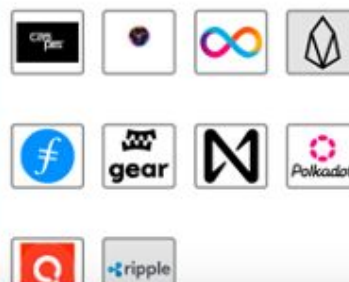
Orchestration & Management



Hosted Platforms



Decentralized Platforms



Debugging & Observability



WebAssembly (WASM) and OpenShift: A Powerful Duo for Modern Applications

November 15, 2023 | by Gaurav Singh, Mrunal Patel and Sohan Kunkerkar

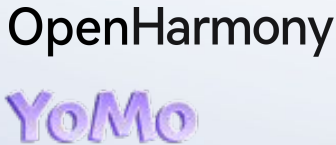
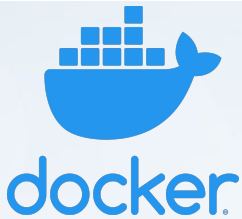
Article

Unlock WebAssembly workloads with Podman on macOS and Windows

December 6, 2023 [Containers, Linux](#)

Florent Benoit
Principal Software Engineer

Build, Share, and Run WebAssembly Apps Using Docker



为什么选择本地部署大模型

实现更自由、更个性化的大模型应用。当前的在线 AI 模型，如 ChatGPT、Bard、Grok 等，虽然功能强大，但受限于商业和技术壁垒，例如搜索引擎的限制、对特定内容的访问限制。

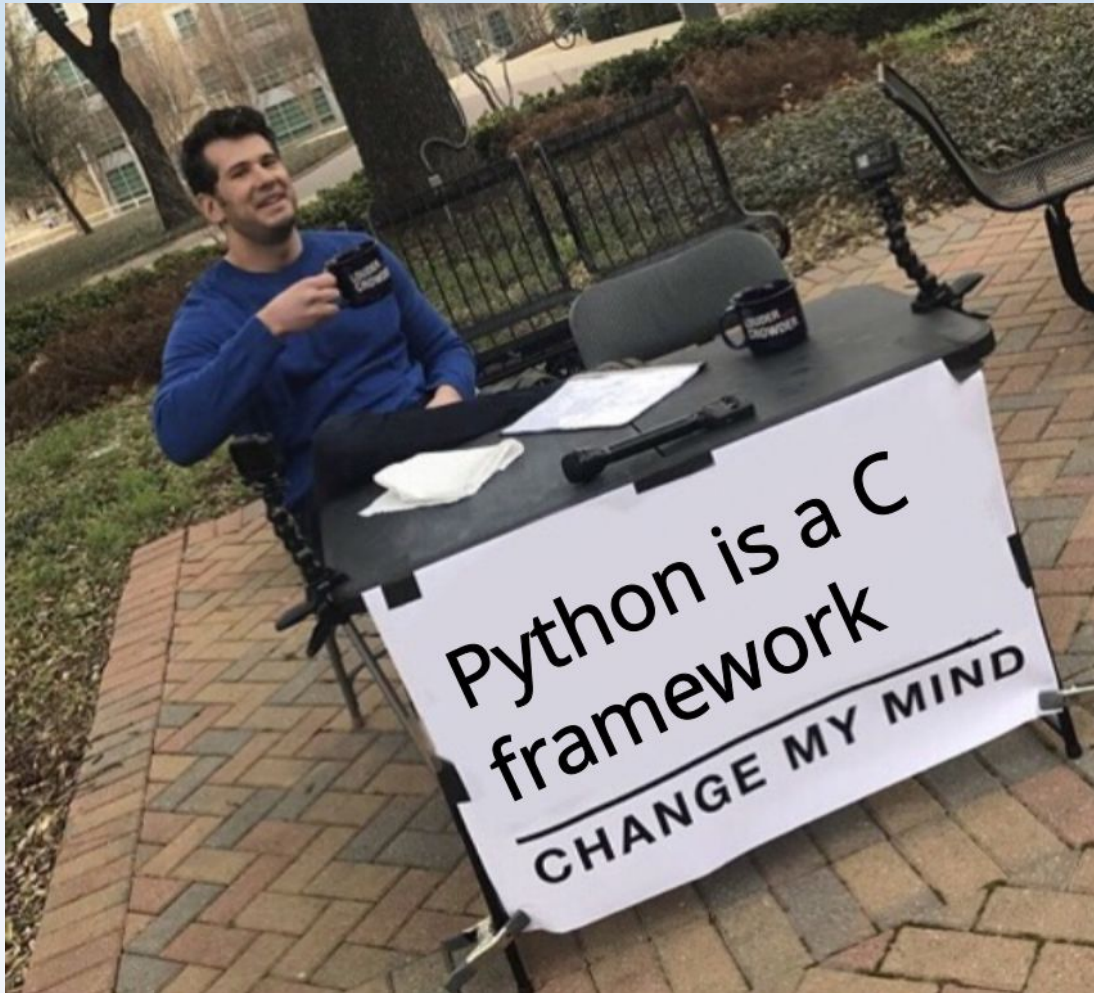
- **在线模型局限：** ChatGPT 只能用 Bing 搜索，Bard 依赖 Google，但遵守 Robots.txt 限制。
- **垄断影响：** 巨头之间的商业竞争限制了 AI 服务的潜力，导致信息孤岛。

1. * Run LLMs on my own Mac fast and efficient Only 2 MBs (secondstate.io)
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218 points by PaulHoule 11 hours ago | flag | hide | 22 comments
14. ▲ LLMs cannot find reasoning errors, but can correct them (arxiv.org)
213 points by koie 13 hours ago | flag | hide | 101 comments
15. * Self-Hosting Open Source LLMs: Cross Devices and Local Deployment of Mistral 7B (secondstate.io)
66 points by 3Sophons 6 hours ago | hide | 15 comments
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为什么要用 Wasm?
Look ma, no Python!







pytorch/pytorch

Sponsored OSSPulls 5M+

By [PyTorch](#) • Updated 5 months ago

PyTorch is a deep learning framework that puts Python first.

Image

OverviewTags

Sort by NewestFilter Tags

TAG latest Last pushed 5 months ago by pytorchbot	DIGEST 82e0d379a5de	OS/ARCH linux/amd64	COMPRESSED SIZE 2.93 GB	docker pull pytorch/pytorch:lat...
TAG 2.0.1-cuda11.7-cudnn8-devel Last pushed 5 months ago by pytorchbot	DIGEST 4f66166dd757	OS/ARCH linux/amd64	COMPRESSED SIZE 6.37 GB	docker pull pytorch/pytorch:2.0...
TAG 2.0.1-cuda11.7-cudnn8-runtime Last pushed 5 months ago by pytorchbot	DIGEST 82e0d379a5de	OS/ARCH linux/amd64	COMPRESSED SIZE 2.93 GB	docker pull pytorch/pytorch:2.0...

Rust + Wasm is a better and much lighter C framework

Rust!



Bojan Tunguz  @tunguz · Apr 21

AGI will be built with Python.

...

Let that sink in.



519



382



5,084



3.8M



Elon Musk  

@elonmusk

...

Rust

4:40 AM · Apr 22, 2023 · **3.7M** Views

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Rust + Wasm

与Python 相比

- 简洁
 - 独立的二进制 app
 - 没有复杂的依赖
 - 跨平台
 - 随时可以云部署
 - 易于更改
- 0.1% of the overall size
- 计算速度提高 100,00 倍

与 C++ 相比

- 安全的容器化应用程序
 - 随时可云部署
- 单一跨平台二进制文件
 - 自动利用本地硬件加速器
- Modern languages
 - Rust, Go, Zig 等等



四个命令行，运行跨平台大语言模型推理app

What is Wasm?

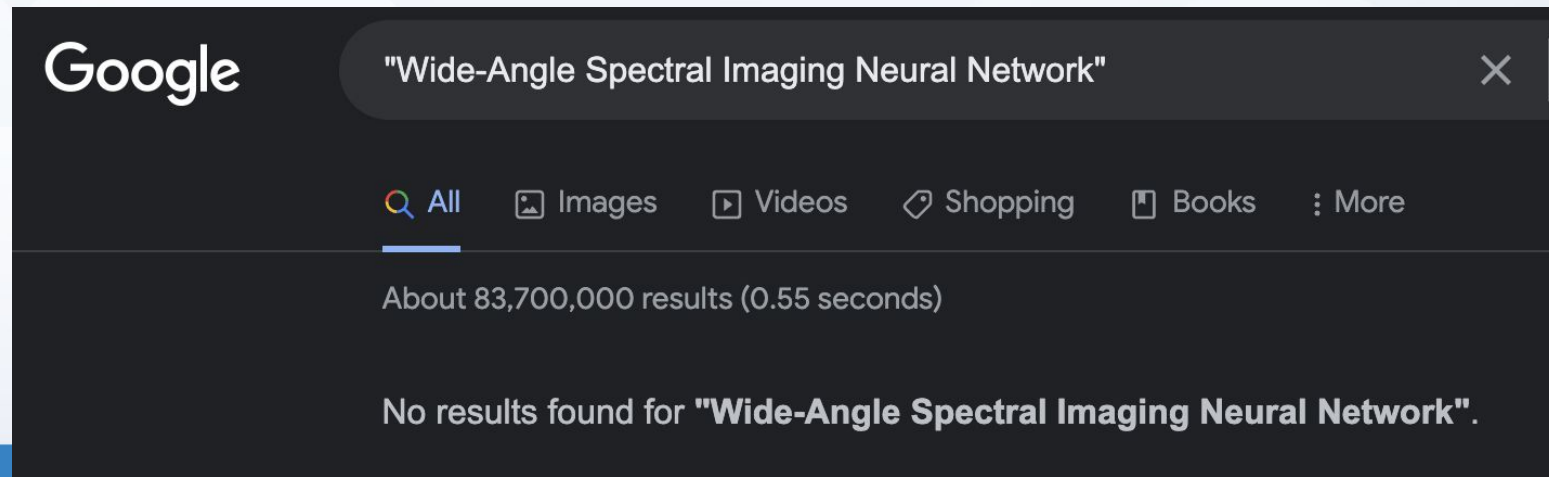
Wasm (WebAssembly) is a binary instruction format for the web that allows developers to compile languages like C, C++, and Rust into compact, efficient, and safe code that can run in web browsers and other environments. It aims to provide a platform-agnostic, portable, and secure way to deploy web applications, improving performance and reducing the size of web pages.

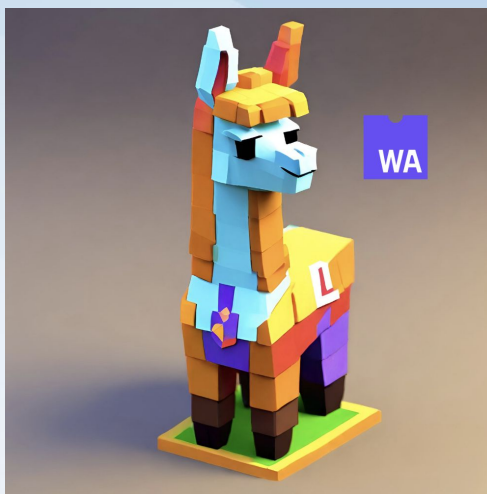
– Llama2 7b model

What is WASI-NN?

WASI-NN stands for "Wide-Angle Spectral Imaging Neural Network." It is a type of neural network architecture that is specifically designed for image recognition tasks, particularly in the field of remote sensing and Earth observation. WASI-NN uses a combination of convolutional and recurrent neural networks to process images and extract useful information, such as land cover classification, crop yield prediction, and object detection.

– Llama2 7b model





快来看Llawwa!



第 1 步:在设备上安装 WasmEdge

```
curl -sSf  
https://raw.githubusercontent.com/WasmEdge/WasmEdge/master/utils/install.sh  
| bash -s -- --plugin wasi_nn-ggml
```

Just one line of command OR you could download and copy the WasmEdge install files manually.

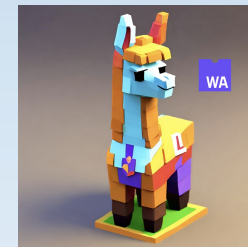
<https://wasmedge.org/docs/start/install>

第 2 步：复制推理应用程序

```
curl -LO  
https://github.com/second-state/llama-utils/raw/main/chat/llama-chat.wasm
```

跨平台的二进制文件。没有其他依赖项。

第 3 步: 下载 Llaw LLM



```
curl -LO  
https://huggingface.co/juntaoyuan/llawa/resolve/main/llama-2-7b-chat-wasm-q5_k_m.gguf
```

Models 2332 🔍

⬆️ Sort: Recently Updated

TheBloke/calml2-7B-chat-AWQ 🔗 Text Generation • Updated about 1 hour ago	TheBloke/stockmark-13B-GGUF 🔗 Text Generation • Updated about 2 hours ago • ❤️ 1
TheBloke/calml2-7B-chat-GGUF Updated about 2 hours ago • ❤️ 1	TheBloke/calml2-7B-chat-GPTQ Updated about 2 hours ago
TheBloke/deepseek-coder-33B-instruct-GGUF Updated about 2 hours ago • ❤️ 20	TheBloke/deepseek-coder-6.7B-instruct-GGUF Updated about 3 hours ago • ❤️ 5
TheBloke/deepseek-coder-1.3b-instruct-GGUF Updated about 3 hours ago • ❤️ 2	TheBloke/deepseek-coder-6.7B-instruct-AWQ 🔗 Text Generation • Updated about 3 hours ago
TheBloke/deepseek-coder-1.3b-instruct-AWQ 🔗 Text Generation • Updated about 3 hours ago	TheBloke/deepseek-coder-33B-instruct-AWQ 🔗 Text Generation • Updated about 3 hours ago • ⬇️ 23 • ❤️ 5

⌵ Expand 2332 models

同时支持 Llama2, CodeLlama, Llemma, Codeshell, Mistrial, MistrialLite, TinyLlama, Baichuan, BELLE, Alpaca, Vicuna, OpenChat, Starcoder, OpenBuddy, 等等！欢迎社区贡献

成功啦！

```
wasmedge --dir ... --nn-preload default:GGML:AUTO:llama-2-7b-chat-wasm-q5_k_m.gguf  
llama-chat.wasm
```

Options:

- `--stream-stdout`: streaming the responses to the terminal
- `--batch-size`: the batch size of the token processor
- `--log-stat`: provides stats on GPU use and speed
- `--reverse-prompt`: stop generation when this sequence appears

代码

The entire inference code fits into ½ of a PPT slide

JavaScript API, powered by WasmEdge-QuickJS, is coming!

```
// load the model into wasi-nn
let graph =
    wasi_nn::GraphBuilder::new(wasi_nn::GraphEncoding::Ggm1, wasi_nn::ExecutionTarget::AUTO)
        .build_from_cache(&model_name)
        .expect("Failed to load the model");

// initialize the execution context
let mut context = graph
    .init_execution_context()
    .expect("Failed to init context");

// set metadata
let metadata = serde_json::to_string(&options).expect("Fail to serialize options");
context
    .set_input(
        1,
        wasi_nn::TensorType::U8,
        &[1],
        metadata.as_bytes().to_owned(),
    )
    .expect("Fail to set metadata");

// set input tensor
let tensor_data = prompt.as_str().as_bytes().to_vec();
context
    .set_input(0, wasi_nn::TensorType::U8, &[1], &tensor_data)
    .expect("Failed to set prompt as the input tensor");

// execute the inference
context.compute().expect("Failed to complete inference");

// retrieve the output
let mut output_buffer = vec![0u8; *CTX_SIZE.get().unwrap()];
let mut output_size = context
    .get_output(0, &mut output_buffer)
    .expect("Failed to get output tensor");
output_size = std::cmp::min(*CTX_SIZE.get().unwrap(), output_size);
let output = String::from_utf8_lossy(&output_buffer[..output_size]).to_string();
```



创建一个兼容OpenAI 的API服务器

创建一个云服务

```
wasmedge --dir ... --nn-preload default:GGML:AUTO:llama-2-7b-chat-wasm-q5_k_m.gguf  
llama-api-server.wasm -p llama-2-chat -s 0.0.0.0:8080
```

Try it!

```
curl -X POST http://localhost:8080/v1/chat/completions \  
  -H 'accept: application/json' \  
  -H 'Content-Type: application/json' \  
  -d '{"messages":[{"role":"system", "content": "You are a helpful assistant. Answer  
each question in one sentence."}, {"role":"user", "content": "What is Wasm?"}],  
"model":"llama-2-chat"}'
```

用containerd/Docker/k8s管理

Video demo: <https://www.youtube.com/watch?v=1qjtxZWXh5w>

Try it yourself: <https://github.com/second-state/runwasi#llama2-example>

Limitations:

- Beta software not yet merged upstream to runwasi yet
- CPU only at this moment

在边缘设备上使用 WasmEdge 和 YoMo 对实时数据流进行 AI 推理，以实现高性能、小体积和 AI 支持。





ggml



YOLOv5

接下来

- 用 chatbot-ui 框架创建一个聊天机器人
 - <https://second-state.github.io/chatbot-ui/>
- 用 LangChain or LlamaIndex 创建一个 RAG app
 - <https://www.langchain.com/>
 - <https://www.llamaindex.ai/>
- 用 flows.network 创建 RAG Discord, Telegram, Slack 聊天机器人
 - <https://flows.network/> <- This uses Rust and Wasm!

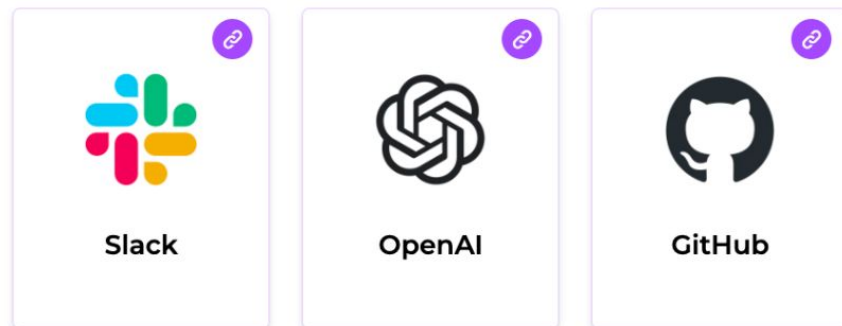


实际应用场景 - 旅行计划

- **场景描述:** 在规划旅行时, 用户需要综合多个平台的信息, 如小红书的旅游攻略、携程的机票和酒店信息、百度地图的路线规划等。
- **问题:** 单一的在线 AI 平台无法提供全面的服务, 因为它们不能跨平台获取和整合信息。

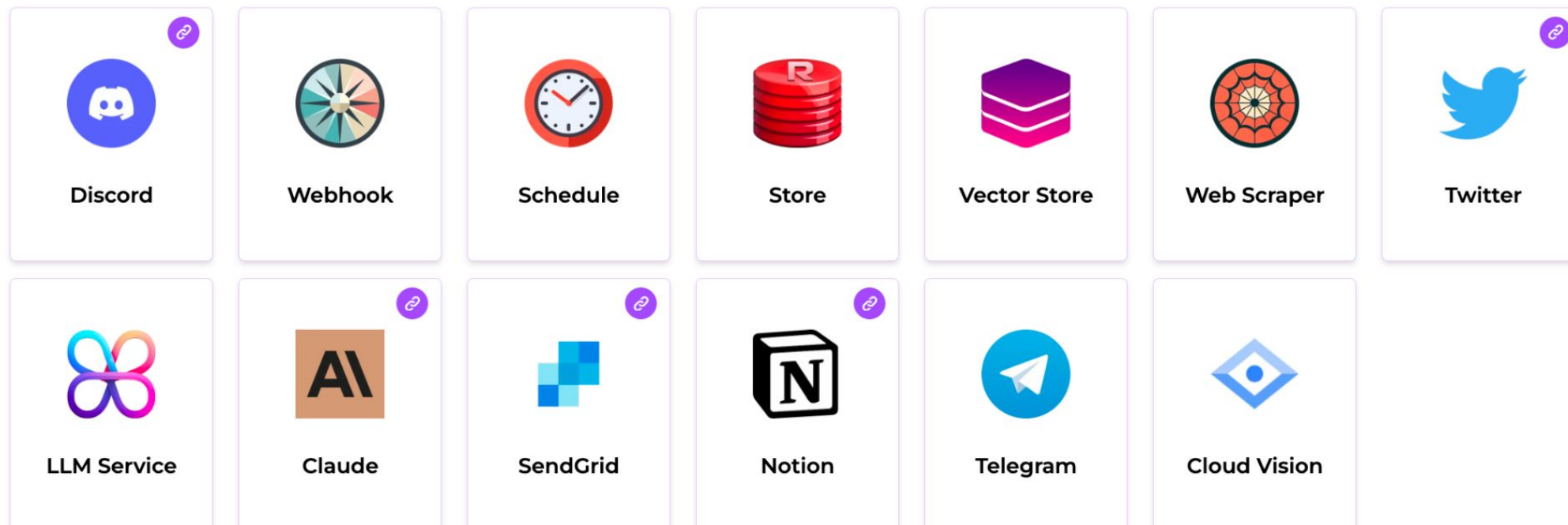
Integrations

Connected



实现为API并提供了
Rust的SDK进行调用

Others



Flows.network: 用Rust编写
一个LLM应用
-<https://zhuanlan.zhihu.com/p/667494969>





WasmEdgeRuntime



CLOUD NATIVE
COMPUTING FOUNDATION

提issue+保持联系！



<https://github.com/WasmEdge/WasmEdge>

Connect with me GitHub/Twitter: @juntao @mileyfu

Join the [WasmEdge discord](https://discord.com/invite/U4B5sFTkFc): <https://discord.com/invite/U4B5sFTkFc>



一键在本地或边缘运行开源 LLM

同时可创建兼容 OPEN AI 的 API 服务器

在终端上运行以下命令行↓

运行
llm.sh

```
bash <(curl -sSfL 'https://code.flows.network/we  
bhook/iwYN1SdN3AmPgR5ao5Gt/run-llm.sh')
```

一键部署大模型，可跨平台在 Linux、macOS、x86、arm、Apple Silicon 和 NVIDIA GPU 上运行 LLM 聊天应用

支持 GGUF 格式的 Llama2 系列所有大模型

由 Rust 和 WasmEdge (CNCF沙箱项目) 赋能



<https://www.secondstate.io/run-llm/>



谢谢大家

